Technical Service Bulletin

Technical Service Bulletin: TSB180069	**Released Date:** 24-May-2022
Lubricating System Cleaning	

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected:

- HSK78G All Versions
- K19
- K38
- K50
- QSK19 All Versions
- QSK23 All Versions
- QSK38 All Versions
- QSK45 All Versions
- QSK50 All Versions
- QSK60 All Versions
- QSK78 All Versions
- QSK95 All Versions
- QST30 All Versions
- QSV81 All Versions
- QSV91 All Versions
- V28

Issue

This document provides inspection and repair recommendations following a debris generating malfunction. This procedure outlines the best procedure to clean an engine following a debris generating malfunction; however, customer constraints may limit the feasibility of following all recommendations. In these cases, it is suggested that as many of the steps as possible are followed.

If a debris generating malfunction has occurred, the lubricating oil system **must** be inspected and cleaned. Further engine damage may result if the steps outlined in this document are **not** followed.

For any type of debris-generating malfunction, the engine **must** be inspected and repaired as required, taking care to inspect associated hardware that either directly supplies lubricating oil to the failed component, or that receives lubricating oil from the failed component, as debris produced in the malfunction may have caused damage to these neighboring components.

General examples of debris generating malfunctions include but are **not** limited to:

1. Modular common rail system (MCRS) fuel pump camshaft or camshaft roller malfunction
2. Dropped exhaust valve or intake valve
3. Connecting rod bearing malfunction
4. Camshaft spalling or skidding
5. Gear train malfunction

Inspection

Cut open lubricating oil filter(s) or open the centrifuge and inspect for debris. If debris is present, complete all cleaning and inspection steps listed below as part of the repair procedure.

1. Drain lubricating oil. See corresponding Service Manual. Reference Procedure 007-037 in Section 7.

2. Remove lubricating oil pan (if possible); clean thoroughly. See corresponding Service Manual. Reference Procedure 007-025 in Section 7.

3. Remove, disassemble, and inspect lubricating oil pump for reuse See corresponding Service Manual. Reference Procedure 007-031 in Section 7.

4. Remove and inspect main bearings and connecting rod bearings as outlined in Table 1. See corresponding Service Manual. Reference Procedure 001-006 in Section 1.

- Inspect main bearings at locations identified in Table 1 below.
- If primary inspected main bearings do **not** meet reuse guidelines, inspect connecting rod bearings as indicated in Table 1. The remaining main bearings **must** be inspected.
- If primary inspected connecting rod bearings do **not** meet reuse guidelines, inspect all remaining connecting rod bearings.
- Table 1 below identifies the first and last bearings to receive lubricating oil based on the engine design. The purpose of the inspection is to review the first and last bearings to receive engine lubricating oil as these tend to accumulate the most debris. Customer constraints may **not** permit inspection of the listed bearing locations; however, all other bearings may **not** provide adequate assessment.
- The K38/50 and QSK38/50 engines can have filtration options mounted on either the right bank or the left bank of the engine. First and last bearings to receive lubricating oil can change based on which bank lubricating oil filters are mounted to.
- Some engines have front or rear main bearings that do **not** feed connecting rod bearings. These bearings are identified in Table 1 below, as well as the first and last bearings that do feed connecting rod bearings.

Table 1, Bearing Inspection by Engine		
Engine	**Main Bearings**	**Connecting Rod Bearings**
QSK19/K19	6, 2 (1**)	5, 1

Table 1, Bearing Inspection by Engine		
Engine	**Main Bearings**	**Connecting Rod Bearings**
QSK23	6, 1	5, 1
V28	7, 1	6L, 1R
QST30	5, 1	4L, 4R, 1L, 1R
QSK38/K38	RBF* 6, 1 LBF* 4, 1	5R, 6L, 1L 3R, 4L, 1L
QSK45	7, 1	6R, 1L
QSK50/K50	RBF* 8, 1 LBF* 6, 1	7R, 8L, 1L 5R, 6L, 1L
QSK60	9, 1	8R, 1L
QSK78	10, 1	9R, 1L
QSV81/QSV91	9 (10**) 1	9L, 9R, 1L, 1R
QSK95	7, 2 (1**)	6L, 6R, 1L, 1R
*RBF – Right Bank Filter; LBF – Left Bank Filter. ** Furthest main bearing that does **not** feed connecting rod bearings.		

5. Inspect crankshaft for reuse at the location of bearing inspections. If crankshaft does **not** meet reuse guidelines, repair accordingly. If crankshaft meets reuse guidelines, replace main and connecting rod bearings as required.

6. Remove, inspect and clean lubricating oil filter head and all associated hardware including any blanking plates (KV engines **only**). See corresponding Service Manual. Reference Procedure 007-015 in Section 7.

If equipped with Eliminator™, proceed with disassembly, inspection, and cleaning. Eliminator™ filtration systems **must** be cleaned and inspected to ensure that filtration screens have **not** been damaged during the malfunction due to accumulation of debris that can cause screen rupture. See corresponding Service Manual. Reference Procedure 007-067 in Section 7.

If equipped with remote-mounted lubricating oil filter head, proceed with disassembly, inspection, and cleaning. See corresponding Service Manual. Reference Procedure 007-017 in Section 7. Replace lubricating oil hoses between engine and filter head. See equipment manufacturer service information.

7. Install spin-on lubricating oil filter head and lubricating oil filters. Spin-on lubricating oil filters are recommended for use during and immediately following debris-generating malfunction repairs as spin-on lubricating filters are more capable of catching debris when flushing the engine lubricating oil system at low volumes and pressures.

8. Remove all valve covers and inspect for signs of debris. See corresponding Service Manual. Reference Procedure 003-011 in Section 3. If debris is found, disassemble and clean camshaft followers and rocker levers. See corresponding Service Manual. Reference Procedure 003-009 in Section 3.

9. Inspect turbocharger for any visible white smoke. See corresponding Service Manual. Reference Procedure 010-033 in Section 10. Check the turbocharger axial and radial clearance Specifications. See corresponding Service Manual. Reference Procedure 010-034 in Section 10.

10. Run brass or plastic brush through oil rifle and lubricating oil cooler drillings if possible. Oil rifle passages are **only** accessible if the engine is removed from chassis as part of the repair.

11. Run fresh oil through oil passages.

12. Discard spin-on lubricating oil filters and lubricating oil.

13. Replace lubricating oil coolers. KV engine lubricating oil flow routes lubricating oil through lubricating oil coolers prior to filtration. This allows for greater levels of debris to be caught in the lubricating oil coolers. See corresponding Service Manual. Reference Procedure 007-007 in Section 7.

14. Install all applicable hardware removed in steps 1-12 (except for Eliminator © filter).

15. Install new spin-on lubricating oil filters.

16. Run engine until operating temperature is reached. Replace lubricating oil and filters (if engine **not** equipped with Eliminator©), or install cleaner Eliminator™ (if equipped).

Document History

Date	Details
2018-5-31	Module Created
2018-12-19	Non-Product Problem Solving (PPS)
2021-7-26	Added number 9 in Inspection section.
2022-5-24	Updated number 6 in Inspection section.

Last Modified: 24-May-2022